General Mathematics — Exercise Sheet 25

The Riemann integral and uniform continuity

Variant: Statements

Exercise 1 (Changing finitely many values). Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be Riemann integrable. Let $g : [a, b] \to \mathbb{R}$ agree with f except possibly on a finite set F. Prove that g is integrable and that $\int_a^b g = \int_a^b f$. Use Lecture 25's Darboux criterion and linearity.

Exercise 2 (An integral with a jump). Fix $a < c < b$ and real numbers p, q, s. A function $f : [a, b] \to \mathbb{R}$ satisfies

$$f(x) = \begin{cases} p, & a \le x < c, \\ s, & x = c, \\ q, & c < x \le b. \end{cases}$$

Prove integrability and compute the integral using the preceding exercise and Lecture 25's constant integral and additivity over adjacent intervals.

Exercise 3 (A uniform choice depends on the domain). Let $f(x) = x^2$.

(a) For each fixed $M > 0$, prove uniform continuity on $[-M, M]$ and give an explicit δ for a prescribed $\varepsilon > 0$.

(b) Prove that f is not uniformly continuous on $\mathbb{R}$ by giving, for every $\delta > 0$, two points less than δ apart whose squared values differ by at least 1.

Use the definition in Lecture 25 and the Archimedean property from Lecture 2.

Exercise 4 (Why sufficiently small mesh works). Let $f : [a, b] \to \mathbb{R}$ be bounded and Riemann integrable, where $a < b$. Prove the following strengthening of the Darboux criterion: for every $\varepsilon > 0$ there is $\delta > 0$ such that every partition Q with $\|Q\| < \delta$ satisfies

$$U(Q, f) - L(Q, f) < \varepsilon.$$

Deduce that every tagged sum $S(Q, f, t) = \sum_i f(t_i)\Delta x_i$, with $t_i \in [x_{i-1}, x_i]$, differs from the integral by less than ε whenever $\|Q\| < \delta$. Use only Lecture 25's definitions, criterion and refinement comparisons.